Week 2 - Pending Confirmation List (IoT)

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Each row says what the firmware currently assumes and what I need confirmed. Nothing here has been guessed into the final design; the current values are drafts in `src/config/config.cpp`.

#	Topic	What the firmware assumes now	Question	Who can confirm
1	Telemetry packet	Draft JSON, version 0.1-DRAFT, field names chosen by me	What are the agreed common fields, names, units and format?	Joint session (all teams)
2	Event message	Same packet carries an `event` field	Is the event message a separate message? Which fields?	Joint session
3	Transport	Not chosen (`PENDING_CONFIRMATION`)	MQTT or HTTP API? Topic or endpoint, QoS, authentication?	Mentor, Full Stack
4	Timestamp	Taken from GNSS, null without a fix	What time source should be used when there is no GPS fix?	Mentor
5	Offline behaviour	Packet is dropped when the network is down	Should packets be buffered and sent later? How many?	Mentor
6	Impact detection	Placeholder: magnitude at or above 4.0 g	Which algorithm and threshold? What data is available?	Mentor
7	SOS button	Simulated input only	Pin, debounce time, long-press time, cancel window?	Mentor, PCB
8	GNSS interface	Parsing only (NMEA text)	UART or I2C for the MAX-M10S? Baud rate, update rate, which sentences?	Mentor, PCB
9	IMU interface	`IMU_IF_PENDING`	I2C or SPI for the LSM6DSOX? Address, range, data rate?	Mentor, PCB
10	A7670C	Stub that logs "would send"	UART pins, PWRKEY timing, APN, SIM, registration steps?	Mentor, PCB
11	Voltage levels	Not handled in firmware	A7670C UART is 1.8 V logic and ESP32-S3 is 3.3 V: level shifter on the PCB?	PCB
12	Modem power	Not handled in firmware	A7670C peaks at 2.8 A, BQ24075 is limited to 1.5 A: how is the modem powered?	PCB
13	Backup power switching	Not handled in firmware	How does the device switch between vehicle supply and the 18650 backup?	PCB
14	Battery reading	`battery_mv` is null	Is the battery voltage measured by the ESP32-S3? ADC pin and divider? Charger status pins?	PCB
15	Pin map	All pins -1	Final pin assignment for all modules	PCB
16	GPS acquire order	Network connects only after the GPS fix	Should network connection start in parallel with GPS acquisition?	Mentor
17	Timeouts	GPS 30 s, network 30 s, event hold 4 s	Which values are wanted?	Mentor
18	FAULT behaviour	No telemetry, restart after 3 s once the fault is	What should FAULT do? Auto restart or wait? Should	Mentor

		cleared	it report the fault?	
19	Early SOS	SOS only changes the device state in OPERATIONAL and DEGRADED	Must SOS also work while acquiring GPS and connecting to the network?	Mentor
20	Reporting interval	5 s normal, 1 s during an event	Wanted reporting intervals?	Mentor, Full Stack
21	Device ID	Fixed string `CRASH_ESP32_01`	How is the device ID provisioned?	Mentor, Full Stack
22	Power figures	Taken from datasheets, not designed	Final voltages, charge current, regulator choice (for example the LMR33630-Q1 version)	PCB